CAP182 SS2 Part D

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Report for STADIOcare readmission prediction project

The purpose of this report is to provide recommendations regarding the most suitable model for STADIOcare. STADIOcare needs to reduce operational costs, and minimise avoidable hospital readmissions through data-driven decision-making. To support this objective, a machine learning project was conducted using the MIMIC-IV clinical data to predict whether a patient would be readmitted within 30 days of discharge. The two models that were developed and evaluated are Logistic regression (model 1) and Random forest (model 2).

1. Best model between model 1 and model 2

Based on the results in Part C, model 1 (Logistic Regression) is the best model for deployment in STADIOcare. The model achieved higher performance across all evaluation metrics compared to model 2 (Random Forest). Logistic Regression achieved an accuracy of 98.32%, precision of 92.35%, recall of 100%, F1-score of 96.02%, and ROC-AUC score of 0.9999. In comparison, Random Forest achieved an accuracy of 86.83%, precision of 62.50%, recall of 88.04%, F1-score of 73.10%, and ROC-AUC score of 0.9463.

The most important metric in this project is recall because the goal is to identify as many high risk patients as possible. Logistic Regression achieved a recall of 100%, meaning that every readmitted patient in the test dataset was correctly identified. This is valuable in a healthcare environment because missed readmissions can lead to increased treatment costs or additional pressure on hospital resources.

Logistic Regression achieved a higher F1-score and ROC AUC score than Random Forest. The excellent ROC-AUC score shows the model is able to do discriminate and differentiate between patients who are likely and unlikely to be readmitted.

I also used A McNemar test to determine whether the observed difference in classification performance between Logistic Regression and Random Forest was statistically significant. The test was conducted using the predictions generated by both models on the same test dataset. This provides a formal statistical comparison of classifier performance beyond standard evaluation metrics

2. How the selected model can be improved

Logistic Regression performed very well, however there are a few opportunities to improve its effectiveness. Firstly, additional clinical information should be added into the model. The current model uses patient demographics, admission information, service history, transfer data, and Diagnosis Related Group (DRG) indicators. If we

include diagnosis codes, laboratory test results, medication information, and procedure records we would create richer representation of patient data and possibly improve the accuracy of the model's predictions.

Monitoring the model and retraining the model regularly should be implemented. The population of patients, hospital operations, and treatment processes evolve with time. Retraining the model using newly collected data will help maintain predictive accuracy and prevent the model from drifting.

Lastly, more advanced feature engineering techniques could be implemented. Features such as the number of chronic conditions, or frequency of emergency admissions could improve performance of the predictions. These features can provide comprehensive view of a patient's journey and improve the model's predictions.

3. How the model will need to be adapted to work with STADIOcare's requested data

The current model was trained using the MIMIC IV clinical data. To successfully deploy the project STADIOcare will need to adapt their own datasets and business environment.

STADIOcare's data environment includes admissions, discharge records, billing systems, service allocation data, workforce data and other healthcare information stored in different systems. These data sources will need to be integrated into a single analytical platform before deploying the model.

The target variable should still be 'Readmitted within 30 days (Yes/No)' but the model should be retrained using STADIOcare's own data to learn patterns that are unique to STADIOcare's patient population.

4. Do the results align with the literature from part A?

The findings align with the literature review in Part A. The research by Almeida et al. found that effective feature engineering and historical healthcare variables improve hospital readmission prediction. This project showed similar outcomes as engineered features like prior admissions and transfer history contributed to strong model performance.

Research conducted by Halac et al. showed that machine learning techniques can be used successfully to predict 30-day readmissions and highlighted the importance of factors such as length of stay. This project also used length of stay information and variables related to admission as the key predictive inputs.

The reviewed literature found that Random Forest and other ensemble methods usually outperform simpler models. In this project Logistic regression performed better than Random forest. A model's performance depends on dataset characteristics, the selected variables, preprocessing approach etc. This reminds us of the importance of evaluating multiple candidate models rather than assuming that a more complex algorithm will automatically produce better results.